

SCIENCE NOTES

Motion and Time Electric Current and Its Effects

Comprehensive Chapter Notes | CBSE / NCERT
15–22 Pages · Diagrams · Key Points · Examples

Chapter 1 : Motion and Time

Chapter 2 : Electric Current and Its Effects

■ Table of Contents

CHAPTER	TOPIC	PAGE
Chapter 1	Motion and Time	3
	1.1 Introduction to Motion	3
	1.2 Types of Motion	4
	1.3 Speed	5
	1.4 Uniform & Non-Uniform Motion	6
	1.5 Distance–Time Graph	7
	1.6 Measurement of Time	8
	1.7 Simple Pendulum	9
	1.8 Formulas & Solved Examples	10
	1.9 Summary Table	11
	1.10 Practice Questions	12
Chapter 2	Electric Current and Its Effects	13
	2.1 Introduction to Electric Current	13
	2.2 Electric Circuit	14
	2.3 Components of a Circuit	15
	2.4 Effects: Heating Effect	16
	2.5 Effects: Magnetic Effect	17
	2.6 Electromagnets	17
	2.7 Electric Bell	18
	2.8 Chemical Effect	19
	2.9 Formulas & Solved Examples	20
	2.10 Summary Table & Questions	21

Motion and Time

Physics | Class 7 | Chapter 1

01

★ KEY POINTS ★

- Motion: An object is said to be in motion when its position changes with time.
- Rest: An object is said to be at rest when its position does NOT change with time.
- Speed = Distance ÷ Time. SI unit of speed is m/s (metres per second).
- Uniform Motion: Equal distances are covered in equal intervals of time.
- Non-Uniform Motion: Unequal distances are covered in equal intervals of time.
- A distance–time graph for uniform motion is a STRAIGHT line.
- A distance–time graph for non-uniform motion is a CURVED line.
- A horizontal distance–time graph indicates the object is at REST.
- Time period of a pendulum depends only on its LENGTH, not on the mass of the bob.
- One oscillation (complete vibration) = A → O → B → O → A (one full swing).
- Units of time: seconds (s), minutes (min), hours (h). 1 min = 60 s, 1 h = 3600 s.
- Speed can be converted: 1 km/h = (1000 m)/(3600 s) = 5/18 m/s.

1.1 Introduction to Motion

In our daily life, we observe many objects moving around us — a running car, a flying bird, water flowing in a river, or the hands of a clock. **Motion** is defined as the change in position of an object with respect to its surroundings and with time.

To decide whether an object is moving or at rest, we always need a **reference point** (also called the point of reference or origin). The motion or rest of an object is always described *relative* to a chosen reference point.

At REST	In MOTION
Book lying on a table	A moving bus
A building	A flying aeroplane
A tree	A swimming fish
A chair	A rolling ball

Scalar and Vector Quantities (Introduction): Speed is a scalar quantity — it has only magnitude (size). Velocity is a vector — it has both magnitude and direction. At this level, we focus on speed.

1.2 Types of Motion

Types of Motion

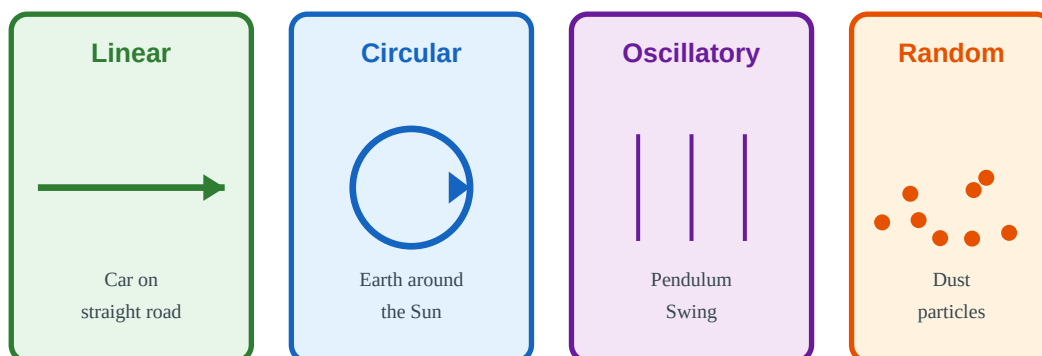


Fig 1.1 — Four main types of motion

Type	Definition	Examples
Linear (Rectilinear)	Motion along a straight line	Car on highway, falling stone
Circular	Motion along a circular path	Earth around Sun, fan blades
Oscillatory / Vibratory	To-and-fro motion about a fixed point	Pendulum, guitar string
Random	No fixed path or direction	Dust particles, Brownian motion

1.3 Speed

Speed is defined as the **distance travelled by an object in unit time**. It tells us how fast or slow an object is moving.

$$\text{Speed} = \text{Distance} \div \text{Time}$$

Or symbolically: $v = d / t$ where v = speed, d = distance, t = time.

Quantity	Symbol	SI Unit	Other Units
Speed	v	m/s	km/h, cm/s
Distance	d	metre (m)	km, cm, mm
Time	t	second (s)	min, h

Unit Conversion: $1 \text{ km/h} = 5/18 \text{ m/s}$. To convert $\text{km/h} \rightarrow \text{m/s}$, multiply by $5/18$. To convert $\text{m/s} \rightarrow \text{km/h}$, multiply by $18/5$.

1.4 Uniform and Non-Uniform Motion

Uniform Motion	Non-Uniform Motion
Equal distances in equal time intervals	Unequal distances in equal time intervals
Speed remains constant	Speed changes (increases or decreases)
Distance–time graph: straight line	Distance–time graph: curved line
Example: Light travels in vacuum	Example: Car in city traffic

Example: Earth's revolution (approx.)

Example: A ball thrown upward

Average Speed: When the speed of an object varies during its journey, we use *average speed*.

$$\text{Average Speed} = \text{Total Distance Travelled} \div \text{Total Time Taken}$$

1.5 Distance–Time Graph

A **distance–time graph** is a graphical representation of the motion of an object. Time is plotted on the X-axis and Distance is plotted on the Y-axis.

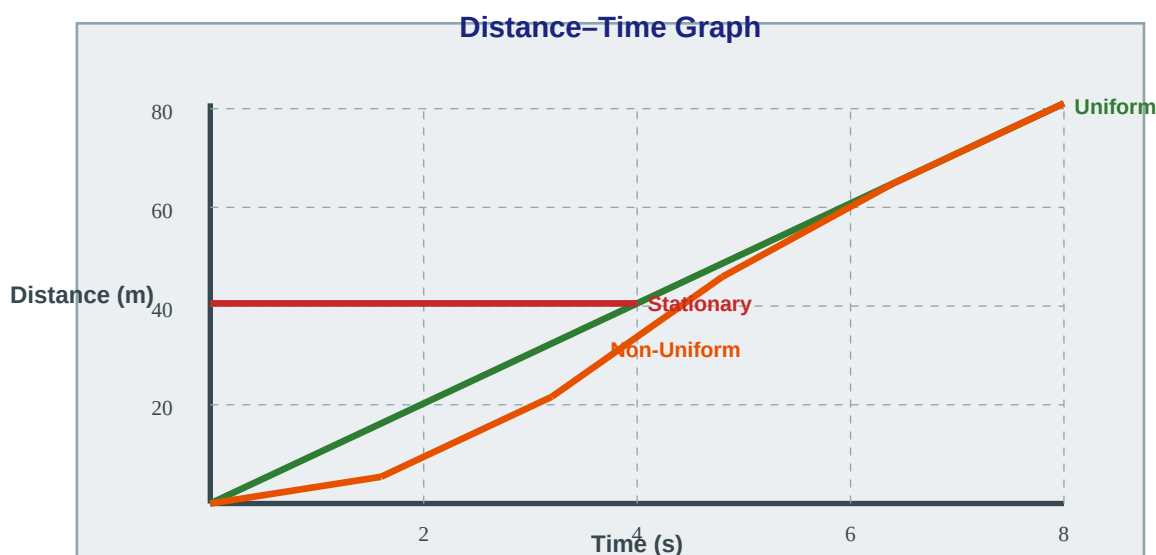


Fig 1.2 — Distance–Time graph showing uniform, non-uniform, and stationary cases

Graph Shape	Type of Motion	Speed
Straight line (slanted)	Uniform motion	Constant
Curved line (steeper)	Accelerating (speeding up)	Increasing
Curved line (flatter)	Decelerating (slowing down)	Decreasing
Horizontal straight line	Object at rest (stationary)	Zero

Key Observation: The steeper the slope of the distance–time graph, the greater the speed of the object.
The slope = Rise/Run = Distance/Time = Speed.

1.6 Measurement of Time

Time is measured using clocks, watches, stopwatches, and sundials. The SI unit of time is the **second (s)**. Other common units include minutes (min) and hours (h).

Conversion Table:

Unit	Symbol	Equivalent
Second	s	Base unit
Minute	min	1 min = 60 s
Hour	h	1 h = 60 min = 3600 s
Day	d	1 d = 24 h = 86400 s

Millisecond	ms	1 ms = 0.001 s
Microsecond	μs	1 μs = 0.000001 s

In ancient times, time was measured by shadow clocks (sundials), water clocks (clepsydra), sandglasses, and burning candles. Modern clocks use the vibrations of a quartz crystal or atomic oscillations for extreme accuracy.

1.7 Simple Pendulum

What is a Pendulum?

A simple pendulum consists of a small heavy ball (called the **bob**) tied to a light, inextensible string, suspended from a rigid support. When displaced from its mean position and released, it swings to-and-fro.

Key Terms:

- **Mean Position (O):** The equilibrium position (rest position).
- **Extreme Positions (A, B):** The farthest points of swing.
- **Oscillation:** One complete to-and-fro motion ($A \rightarrow O \rightarrow B \rightarrow O \rightarrow A$).
- **Time Period (T):** Time taken for one complete oscillation.
- **Amplitude:** Maximum displacement from mean position.

Simple Pendulum

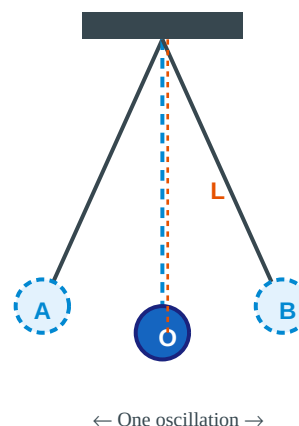


Fig 1.3 — Simple pendulum showing oscillation

Time Period Formula:

$$T = 2\pi \sqrt{L/g}$$

Where L = length of pendulum, g = acceleration due to gravity (9.8 m/s^2). Note: T is **independent of the mass of the bob**.

Factors affecting Time Period: Only the **length (L)** of the pendulum affects the time period. Longer pendulums have a longer time period (swing more slowly). The mass of the bob and the angle of swing (for small angles) do NOT affect T .

1.8 Important Formulas and Solved Examples

Formula	Rearranged Forms	Use
Speed (v) = d / t	$d = v \times t$ $t = d / v$	Find speed, distance, or time
Average Speed = Total d / Total t —		For varying speeds
1 km/h = 5/18 m/s	1 m/s = 18/5 km/h	Unit conversion

Q: A car travels 150 km in 3 hours. Find its speed in km/h and m/s.

Example 1 — Finding Speed

Question: A car travels 150 km in 3 hours. Find its speed in km/h and m/s.

Solution:

- Given: Distance = 150 km, Time = 3 h
- Speed = Distance ÷ Time = $150 \div 3 = 50$ km/h
- Converting: $50 \times 5/18 = 250/18 \approx 13.89$ m/s

Q: A cyclist rides at 20 km/h for 2.5 hours. How far does he travel?

Example 2 — Finding Distance

Question: A cyclist rides at 20 km/h for 2.5 hours. How far does he travel?

Solution:

- Given: Speed = 20 km/h, Time = 2.5 h
- Distance = Speed × Time = $20 \times 2.5 = 50$ km

Q: How long will a train take to cover 360 km at a speed of 90 km/h?

Example 3 — Finding Time

Question: How long will a train take to cover 360 km at a speed of 90 km/h?

Solution:

- Given: Distance = 360 km, Speed = 90 km/h
- Time = Distance ÷ Speed = $360 \div 90 = 4$ hours

Q: A bus covers 60 km in the first hour and 80 km in the next 2 hours. Find average speed.

Example 4 — Average Speed

Question: A bus covers 60 km in the first hour and 80 km in the next 2 hours. Find average speed.

Solution:

- Total Distance = $60 + 80 = 140$ km
- Total Time = $1 + 2 = 3$ h
- Average Speed = $140 \div 3 \approx 46.67$ km/h

Q: Convert 72 km/h to m/s.

Example 5 — Speed Conversion

Question: Convert 72 km/h to m/s.

Solution:

- $72 \text{ km/h} \times 5/18 = 360/18 = 20$ m/s

1.9 Chapter Summary Table

Concept	Definition	Key Fact
---------	------------	----------

Motion	Change in position w.r.t. time	Needs a reference point
Rest	No change in position	Relative to reference
Speed	Distance per unit time	SI unit: m/s
Uniform Motion	Constant speed, straight path	d-t graph: straight line
Non-Uniform Motion	Varying speed	d-t graph: curved line
Distance–Time Graph	Graphical motion representation	Slope = Speed
Pendulum Period	Time for one oscillation	Depends only on length
Time units	s, min, h	1 h = 3600 s

1.10 Practice Questions

- 1. Define motion and give two examples from daily life.
- 2. What is the SI unit of speed? Write the formula for speed.
- 3. Differentiate between uniform and non-uniform motion with examples.
- 4. What does a horizontal line in a distance–time graph indicate?
- 5. A train covers 400 km in 5 hours. Calculate its speed in m/s.
- 6. Convert 54 km/h into m/s.
- 7. What is the time period of a pendulum? On what factors does it depend?
- 8. A pendulum completes 20 oscillations in 40 seconds. Find its time period.
- 9. An object covers 30 m in 6 s and 80 m in the next 8 s. Find its average speed.
- 10. Draw a distance–time graph for an object moving at constant speed.

★ Higher-Order Questions:

- 11. Why is the motion of the Earth's revolution considered approximately uniform?
- 12. A car's speed is 72 km/h. If it brakes and stops in 10 seconds, describe the d–t graph for this deceleration.
- 13. Two pendulums have lengths 25 cm and 100 cm. Which has a longer time period and why?

Electric Current and Its Effects

Physics | Class 7 | Chapter 2

02

★ KEY POINTS ★

- Electric current is the flow of electric charge through a conductor.
- A circuit must be CLOSED (complete) for current to flow.
- Components: Cell/Battery, Switch, Bulb, Connecting wires.
- Heating Effect: Current through a resistance produces heat. (e.g., electric heater, iron).
- The filament of an electric bulb glows because of the heating effect.
- Magnetic Effect: A current-carrying conductor produces a magnetic field around it.
- An Electromagnet is a temporary magnet created by winding a coil around an iron core.
- Strength of an electromagnet increases with more turns of wire or higher current.
- Electric Bell works on the magnetic effect of electric current.
- Chemical Effect: Current passed through a liquid may cause chemical changes (electrolysis)...
- Good conductors allow current to flow; insulators do NOT allow current to flow.
- Safety: Never touch exposed wires; always use fused circuits; MCBs protect circuits.

2.1 Introduction to Electric Current

Electric current is the **flow of electrons** (negative charges) through a conductor. By convention, the direction of current is taken as opposite to the direction of electron flow — from the **positive terminal** to the **negative terminal** of the cell (conventional current).

A **conductor** is a material that allows electricity to pass through it (e.g., copper, iron, silver). An **insulator** does not allow electricity (e.g., rubber, wood, plastic).

Conductors	Insulators
Copper	Rubber
Silver	Wood
Aluminium	Plastic
Iron	Glass
Graphite	Dry air

2.2 Electric Circuit

An **electric circuit** is a closed path along which electric current flows. The current flows from the positive terminal of the battery, through the external circuit, and back to the negative terminal.

Simple Electric Circuit (Series)

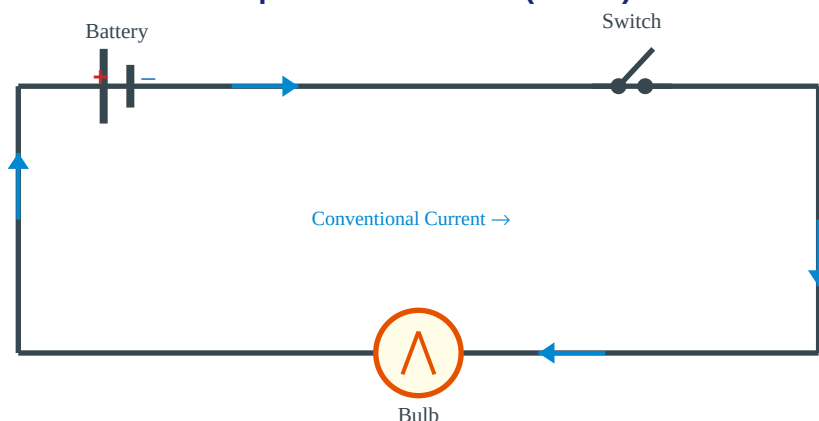


Fig 2.1 — Simple series circuit with battery, switch, and bulb

Open Circuit	Closed Circuit
Circuit is incomplete / broken	Circuit is complete / connected
Current does NOT flow	Current flows freely
Switch is open (OFF)	Switch is closed (ON)
Devices do NOT work	Devices work normally

2.3 Components of an Electric Circuit

Component	Symbol (description)	Function
Cell	— — (long & short line)	Provides electric energy (EMF)
Battery	Multiple cells	Combination of cells for more voltage
Switch	Open gap in wire	Opens/closes the circuit
Bulb	Circle with filament	Converts electrical energy to light
Ammeter	A inside circle	Measures current (connected in series)
Voltmeter	V inside circle	Measures voltage (connected in parallel)
Resistor	Rectangle	Opposes the flow of current
Connecting wire	Plain line	Carries current between components
Fuse	Short thin wire	Safety device — melts if excess current flows

Series vs Parallel Circuits: In a series circuit components are connected end-to-end in a single loop — if one fails, all stop. In a parallel circuit, components are connected across the same two points — if one fails, others still work (used in household wiring).

2.4 Effect of Electric Current — Heating Effect

When electric current flows through a conductor, the conductor becomes hot. This is called the **Heating Effect (or Joule's Effect) of electric current**. The heat produced depends on:

- The **current (I)** — more current → more heat.

- The **resistance (R)** — higher resistance → more heat.
- The **time (t)** — longer time → more heat.

Heat Produced $H = I^2 \times R \times t$ (Joule's Law)

Heating Effect of Current

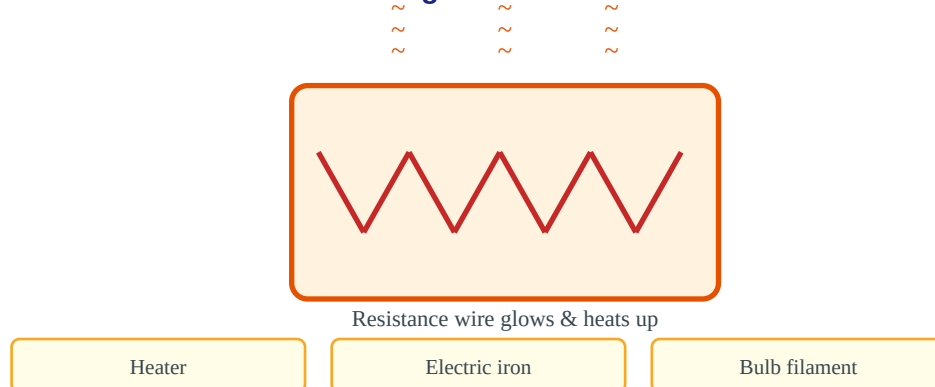


Fig 2.2 — Heating effect: resistance wire heats up

Applications of Heating Effect:

Electric Heater / Room Heater: Nichrome wire coil has high resistance; heats up and warms the room.

Electric Iron: Soleplate gets heated; used to press/iron clothes.

Electric Bulb (Incandescent): Thin tungsten filament glows white-hot and emits light.

Electric Toaster / Kettle: Resistance coil heats the bread or water.

Electric Fuse: A thin wire that melts and breaks the circuit when current is too high — protects appliances.

■ **SAFETY NOTE:** Overloading a circuit (using too many appliances) causes excess heat, which may start a fire. Always use fuses or MCBs to protect circuits!

2.5 Effect of Electric Current — Magnetic Effect

When current flows through a wire, a **magnetic field** is produced around it. This was first discovered by **Hans Christian Oersted** in 1820, when he noticed a compass needle deflecting near a current-carrying wire.

Key observations of Oersted's experiment:

- The compass needle deflects when current flows through the wire.
- When the current is reversed, the needle deflects in the opposite direction.
- The stronger the current, the greater the deflection of the needle.
- When current is switched off, the needle returns to its original position.

2.6 Electromagnets

An **electromagnet** is a temporary magnet made by winding an insulated conducting wire (coil) around an iron core and passing current through it. It behaves like a magnet only when current is flowing.

Electromagnet

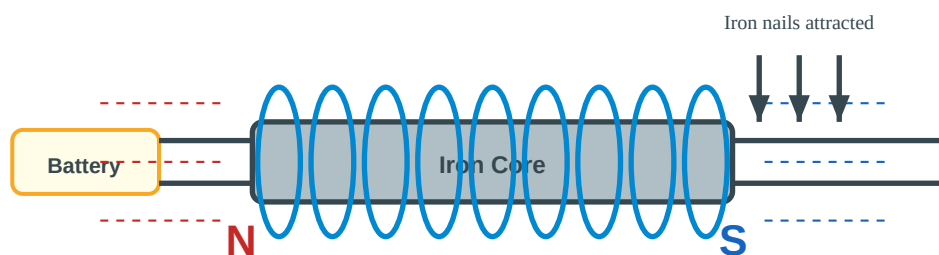


Fig 2.3 — Electromagnet: coil around iron core with N and S poles

Factor	Effect on Electromagnet Strength
Number of turns in coil	More turns → Stronger electromagnet
Amount of current	More current → Stronger electromagnet
Type of core	Soft iron core → Stronger (magnetically soft)
Length of coil	Shorter & more compact → Stronger

Applications of Electromagnets:

- Electric bell, Electric crane (lifting iron scrap)
- Loudspeakers and headphones
- MRI machines in hospitals
- Electric motors and generators
- Magnetic door locks
- Separation of iron from waste/garbage

Electromagnet vs Permanent Magnet:

Electromagnet	Permanent Magnet
Temporary — magnetism can be switched on/off	Permanent — always magnetic
Strength can be varied	Fixed strength
Made of soft iron	Made of hard steel
Used in cranes, bells, motors	Used in compasses, fridge magnets

2.7 Electric Bell

An **electric bell** is a device that uses the magnetic effect of electric current to produce sound. It works on the principle that an electromagnet can attract an iron strip (armature) repeatedly to strike a gong.

Electric Bell – Working

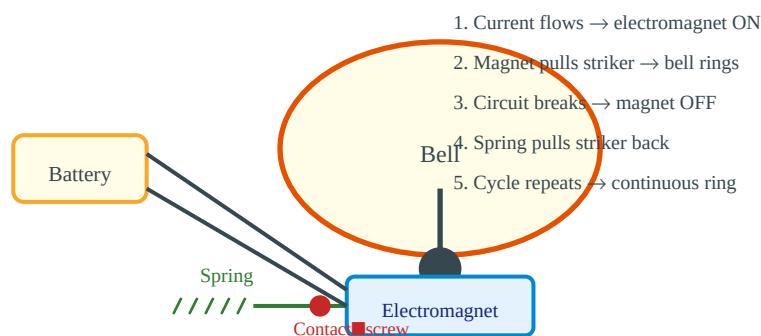


Fig 2.4 — Parts and working of an electric bell

Parts of an Electric Bell:

Electromagnet: An iron core wound with coil — creates magnetic field.

Armature (Spring Strip): A flexible strip of iron that can vibrate.

Striker (Hammer): Attached to armature; strikes the gong.

Gong (Bell): Produces sound when struck.

Contact Screw: A screw that acts as a contact point — makes or breaks circuit.

Battery: Provides electrical energy to the system.

Working of Electric Bell (Step-by-step):

- 1. When switch is pressed, current flows through the circuit.
- 2. Current flows through the electromagnet coil → coil becomes a magnet.
- 3. Electromagnet attracts the armature (iron strip) toward it.
- 4. Striker attached to armature hits the gong → DING sound produced.
- 5. When armature moves toward the magnet, it breaks contact with the contact screw.
- 6. Circuit breaks → electromagnet loses magnetism.
- 7. Spring pulls armature back to original position.
- 8. Contact screw is touched again → circuit is complete again.
- 9. The process repeats rapidly → continuous ringing as long as switch is pressed.

2.8 Chemical Effect of Electric Current

When electric current is passed through certain liquids (called **electrolytes**), chemical changes take place. This is called the **chemical effect** of electric current. The process is called **electrolysis**.

Electrolysis Setup: Two metal plates (electrodes) are dipped in a conducting liquid (electrolyte). When current is passed, one electrode (anode — positive) loses material and the other (cathode — negative) gains it.

Concept	Meaning
Electrolysis	Decomposition of a substance by electric current
Electrolyte	Liquid that allows current and undergoes chemical change (e.g., CuSO_4)

Anode (+)	Positive electrode — loses material (oxidation)
Cathode (–)	Negative electrode — gains material (reduction / deposition)
Electroplating	Coating a metal with another metal using electrolysis

Electroplating:

Electroplating is the process of depositing a thin layer of one metal over another using electrolysis. It is used to:

- Protect metals from rusting (e.g., iron plated with zinc = galvanisation).
- Improve appearance (e.g., silver/gold plating on jewellery).
- Reduce friction (e.g., chromium plating on car parts).
- Prevent corrosion (e.g., tin-plated iron for food cans).

LED (Light Emitting Diode): LEDs are semiconductor devices that emit light when current passes through them. They use the electronic/chemical effect. LEDs are much more energy efficient than incandescent bulbs.

2.9 Important Formulas and Solved Examples

Formula	Meaning
$I = Q / t$	Current = Charge / Time ($A = C/s$)
$V = I \times R$	Ohm's Law: Voltage = Current $\times$ Resistance
$H = I^2 \times R \times t$	Joule's Heating Law
$P = V \times I$	Power = Voltage $\times$ Current (Watts)

Example 1 — Ohm's Law

Q: A bulb has resistance $100\ \Omega$ and a voltage of $200\ V$ is applied. Find the current.

Solution:

- Given: $V = 200\ V$, $R = 100\ \Omega$
- Using Ohm's Law: $I = V / R = 200 / 100 = 2\ A$

Example 2 — Heating Effect

Q: Find the heat produced in a resistance of $5\ \Omega$ when a current of $2\ A$ flows for $10\ s$.

Solution:

- Given: $I = 2\ A$, $R = 5\ \Omega$, $t = 10\ s$
- $H = I^2 \times R \times t = 4 \times 5 \times 10 = 200\ J$

Example 3 — Electromagnet

Q: A coil has 50 turns. If the turns are doubled, how does the electromagnet strength change?

Solution:

- More turns $\rightarrow$ stronger electromagnet.

► Doubling to 100 turns → approximately doubles the magnetic field strength.

Example 4 — Circuit

Q: A torch has 2 cells of 1.5 V each connected in series. Find the total voltage.

Solution:

► In series: Total $V = V_1 + V_2 = 1.5 + 1.5 = 3 \text{ V}$

2.10 Chapter Summary Table

Effect	Cause	Applications
Heating Effect	Current $\times$ Resistance $\rightarrow$ Heat	Heater, Iron, Fuse, Bulb
Magnetic Effect	Current-carrying conductor $\rightarrow$ Magnetic field	Electromagnet, Bell, Motor
Chemical Effect	Current through electrolyte $\rightarrow$ Chemical change	Electroplating, Electrolysis

Practice Questions — Chapter 2

- 1. What is electric current? What is its SI unit?
- 2. Differentiate between a conductor and an insulator with examples.
- 3. What is the heating effect of electric current? Name three appliances based on it.
- 4. Why does the filament of an electric bulb glow, but the connecting wires do not?
- 5. Who discovered the magnetic effect of electric current and when?
- 6. Define an electromagnet. How does it differ from a permanent magnet?
- 7. State any three ways to increase the strength of an electromagnet.
- 8. With the help of a diagram, explain the working of an electric bell.
- 9. What is electroplating? Give two uses of electroplating.
- 10. A current of 3 A flows through a resistance of 4Ω for 5 seconds. Find heat produced.

★ Higher-Order Questions:

- 11. Explain why electric fuse wires are made of materials with a low melting point.
- 12. Why is soft iron used as the core of an electromagnet instead of steel?
- 13. Why are household appliances connected in parallel rather than in series?
- 14. A student builds an electromagnet with 20 turns. Another builds one with 60 turns, both using the same current. Compare their strengths and explain.

■ QUICK REVISION CARD — Both Chapters

Chapter 1: Motion & Time	Chapter 2: Electric Current
Motion = change in position w.r.t time	Current = flow of charge through conductor
Speed = Distance / Time (SI: m/s)	Effects: Heating, Magnetic, Chemical
Uniform motion $\rightarrow$ straight d-t graph	Heating: $H = I^2 R t$ (Joule's Law)

Non-uniform → curved d-t graph	Magnetic: Oersted → deflects compass
Stationary → horizontal d-t graph	Electromagnet: coil + iron core + current
Pendulum T depends only on length	Electric Bell works on magnetic effect
1 km/h = 5/18 m/s	Electroplating uses chemical effect
Average speed = total d / total t	Series: same current; Parallel: same voltage

- ◆ Good Luck with your exam! Remember: understand concepts, don't just memorise.
- ◆ Always draw neat, labelled diagrams in exams — they carry separate marks!
- ◆ Practice numerical problems daily for best results.
- ◆ These notes cover CBSE / NCERT Class 7 Science curriculum.